

Individual Fairness

(1) Details of the analysis structure

I audited the baseline KNN model using a local individual fairness (IF) test on the fixed test set. I computed predicted probabilities and represented each test instance in the model's preprocessed non-sensitive feature space (same space used for prediction), then built a local nearest-neighbor graph (`k_local = 15`) so each person was compared only to similar individuals. For each local pair, I measured an IF violation when the prediction-probability gap exceeded the pair's feature-space distance. I then summarized the same IF metrics at three levels: global, subgroup (`marital-status`, `sex`), and intersection (`marital-status x sex`), while tracking pair coverage (`n_pairs`) to judge reliability of subgroup estimates. This IF protocol was used as the baseline for later cause-of-unfairness checks.

(2) Quantitative results of the analysis

Global IF summary

Metric	Value
Evaluated pairs	97695
Violation rate	0.260648
Avg violation	0.022848
Max violation	0.773675

Compact subgroup IF summary (selected rows)

Attribute	Subgroup	n_pairs	Viol. rate	Avg viol.
sex	Male	47801	0.282337	0.025391
sex	Female	15811	0.223326	0.017215
marital-status	Divorced	2640	0.318561	0.024547
marital-status	Married-civ-spouse	25074	0.301508	0.028579
marital-status	Never-married	18553	0.152482	0.012216
marital-status	Widowed	358	0.069832	0.004430

Intersectional summaries (full table omitted for space) showed additional variation, e.g., `Divorced - Female` (violation rate 0.359026, `n_pairs`=1479) and `Separated - Male` (violation rate 0.526316, `n_pairs`=19), with some small groups having very low or zero pair coverage.

(3) Observations, inferences and insights from the analysis worth noting

The IF audit shows non-trivial local fairness violations under the chosen similarity-based test, with violations occurring across many local comparisons rather than only rare outliers. The subgroup summaries indicate that local prediction consistency is not uniform across the test population: both `sex` and `marital-status` groups differ in violation rate and average violation. The intersectional view adds useful detail, but some intersection groups have very small `n_pairs`, so extreme values in those rows should be interpreted cautiously. Overall, this IF analysis is most useful as a baseline diagnostic: it identifies where local inconsistencies are concentrated, but it does not by itself explain why they occur, which motivates the later cause-of-unfairness analyses.

Appendix note (optional): Full subgroup and intersection tables are provided in the appendix/supplementary results.